

K-means Clustering (Partitioning Method)

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[Select k]

- k represents no of clusters $k \leq n$; n = data points
- (Each cluster must have atleast one point)
- (A point can only be a part of one cluster only)



[Choose k centroids]

(Initially must be chosen from a data point, later from calculations it can be a data point or a location)

Centroids can be chosen from following way :

- 1> Randomly From data points
- 2> First k data points
- 3> Choosing Bet^a a higher and lower Bound



[Calculate distance from one data point to each centroid
Using Euclidean dist]



[Assign the data point to a cluster which is nearest to a centroid
among multiple centroids]



[Update Centroid location by taking mean of all points
coordinates of that cluster ~~and~~ (not mean dist)]



[From the new Centroid repeat the process of
calculating dist until the ~~ass~~ members of a
cluster does not change]

Note: Highly Sensitive to Outliers, as it can move the mean mean
AWAY !!! (to solve that k medoids is used)

MORE ROBUST
To outliers

K-Medoids Clustering (PAM - Partitioning Around Medoids)

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[Select K number of clusters]

($K \leq n$; K = no of clusters to be formed ; n = data point)

(A point can only be part of one cluster)

Here medoid

is always a data

point even

after calculation

[Choose K medoids (must be from data points)]

(Generally chosen Randomly, first K or between Bounds)

[Calculate distance from one data point to each medoid using Manhattan distance]

[Assign the data point to a cluster which is nearest to a medoid among multiple medoid clusters]

[Calculate Total cost (old)]

(Total cost = Sum of Manhattan dist of each point to its assigned medoid)

[Select a non medoid points and calculate Total cost (new) in similar manner]

[Calculate Swap cost = New total cost - Old total cost]

[if Swap cost < 0 ?]

NO

(Cost dec)

[Accept swap]

(Update the new medoid points)

[Rejects swap]

(Keep old medoid points)

Select another non medoid point and repeat process

NO

[Cannot find a swap with that lowers the total cost]

YES

[Stop]